

1	Course Name:
	Electrical Circuit Analysis
2	Course Code:
	IBB12704
3	Year and Semester Offered:
	Year 1 Semester 2
4	Credit Hours Value:
	4 Credit Hours [4 SLT Credit]
5	Prerequisite(s), if applicable:
	NIL
6	Course Synopsis:
	This course aims to provide students with the fundamental concepts of DC and AC electrical circuits and analyse them using different electrical laws and circuit theorems. Students will acquire knowledge to describe important electrical components used in electrical circuits. At the same time, students will be exposed to the current measuring instruments used to investigate and analyse the behaviour of electrical circuits.
7	Course Learning Outcome:
	Upon completion of the course, students should be able to: 1. Apply knowledge of the basic DC and AC circuit theory 2. Analyze electrical circuits using different circuit theorems 3. Perform AC sinusoidal analysis in steady-state response 4. Construct electrical circuits using laboratory equipment or simulation tools
8	Content Outline:
	Topic
	1. Basic Laws and Circuit Theory
	2. Method of Analysis
	3. Circuit Theorems
	4. Capacitor and Inductor
	5. First Order Circuits
	6. Sinusoids and Phasors
	7. AC Steady-State Analysis